

CAROLYN ZENG

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TECHNICAL SUMMARY

- Data and analytics professional with expertise in **C/C++, Python, R, SQL, JavaScript, Node.js**, and **Linux**, complemented by certifications in **AWS Cloud** and **Microsoft Office**.
- Proven ability to translate complex data into actionable in-depth analytics insights through interactive visualizations, predictive modeling, and machine learning techniques, including time series modeling techniques and regressions.
- Strong collaborative and problem-solving skills developed in academic, volunteer, and professional settings, with passion and dedication to data science

PROFESSIONAL EXPERIENCE

Columbia University SPS

Data Lead & Analyst – Capstone Project (Children’s Museum of Indianapolis)

**MS Applied Analytics, Expected 2026
Fall 2025**

- Developed Gradient Boost model to forecast exhibit popularity using attendance data, foot traffic patterns, and exhibit characteristics, enabling data-driven programming decisions with an R-squared value of 0.52
- Integrated multiple data sources (attendance records, foot traffic sensors, exhibit metadata) to identify key drivers of visitor engagement and optimize gallery space utilization
- Delivered actionable insights through comprehensive EDA and visualization dashboards, revealing exhibit lifecycle patterns and audience preferences to inform strategic planning

Housing Demand Analysis & Presentation

April 2025

- Collaborated with a team of 3 peers to perform data cleansing using housing data
- Developed machine learning models to forecast future demand based on economic and demographic factors for up to 3 months in advance.
- Created interactive data visualization dashboards in Tableau to present key insights to various audiences, including price trends and regional demand shifts.

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July 2025 – Current

- Created and annotated varied datasets as inputs for AI/ML model training, directly contributing to the completion of 10+ projects with a labeling accuracy of 89%.
- Collaborated with team members during projects to resolve data creation issues and adjust data guidelines.
- Provided programming and coding expertise as a member of the team's technical solutions group.

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Library Database and UI

Data Analytics Certificate, In Progress

November – December 2024

- Designed and implemented a relational database in Microsoft Access, structuring tables for books, authors, and genres with normalized relationships to minimize redundancy.
- Developed a user-friendly UI with forms for adding/editing records and a search feature filtering by title, author, or topic, improving lookup efficiency.

ADDITIONAL EXPERIENCE

The Field Museum of Natural History

Discovery Squad

June 2022 – On Leave

- Presented artifacts to 100+ visitors weekly, adapting explanations for diverse audiences (e.g., children vs. researchers).

EDUCATION

Columbia University

MS Applied Analytics | GPA: 3.62 | Expected May 2026

University of Illinois

BS Astronomy | Minors: CS, Physics | GPA: 3.65 | 2024

Research Experience

- "Are Plants Intelligent?" (HORT223, 2022) – Analyzed plant cognition studies.
- "Simulation of Planetary Systems": Modeled particle movements in Python (NumPy/Matplotlib) for astronomy coursework.

Certifications: AWS Certified Cloud Practitioner, MOS: Excel Associate, MOS: Access Expert

Languages: English (Native), Mandarin Chinese (Fluent), German (Intermediate)